EXCELLENCE

CLASS 11 - 2026-27

Physics | Vector 01

Vectors

Feel of a Vector

By - Excellence Programme

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Topics to be covered

01 Log formulae and loudness questions

02 Decaying current

03 Binomial approximation when $1 \gg x$

04 Physical quantity: scalar and vector

05 Writing a vector as magnitude times unit vector

06 Angle between two vectors

07 Types of vectors

Log lock lines

$$\log (x^n) = n \log x$$

$$\log (x \text{ to the } 1/m) = (1/m) \log x$$

$$\log(xy) = \log x + \log y \qquad \log(x/y) = \log x - \log y$$

log_b a = log a / log b, taken in the same base. log 1 = 0. $b^0 = 1$.

Keep the base written. Change of base is a fraction of two logs, not a product.

log base 8 of 16

Find log base 8 of 16.

Solution: log base 8 of 16

Step 1: $16 = 2^4$ and $8 = 2^3$

Step 2: $\log_8(16) = \log(2^4) / \log(2^3) = 4/3$

4/3

Rough work / extra work

log base 27 of 3

Find log base 27 of 3.

Solution: log base 27 of 3

Step 1: $27 = 3^3$, so $\log_{27}(3) = \log(3) / \log(3^3) = 1/3$

1/3

log base 100 of 1000

Find log base 100 of 1000.

Solution: log base 100 of 1000

Step 1: $1000 = 10^3$ and $100 = 10^2$

Step 2: $\log_{100}(1000) = 3/2$

3/2

log10 of 0.0001

Find log10(0.0001).

Solution: log10 of 0.0001

Step 1: $0.0001 = 10^{-4}$

log10(0.0001) = -4

Rough work / extra work

log base 0.01 of 10

Find log base 0.01 of 10.

Solution: log base 0.01 of 10

Step 1: $0.01 = 10^{-2}$, so the base is 10^{-2}

Step 2: log of 10 in base $10^{-2} = 1/(-2) = -1/2$

-1/2

Loudness 50 dB

Loudness is 50 dB. $L = 10 \log_{10}(I/I_0)$ with $I_0 = 10^{-12}$ W/m^2. Find I.

Solution: Loudness 50 dB

Step 1: $50 = 10 \log_{10}(I/I_0)$, so $5 = \log_{10}(I/I_0)$

Step 2: $I/I_0 = 10^5$

Step 3: $I = 10^5 \text{ times } 10^{-12} = 10^{-7}$

$I = 10^{-7}\ W/m^2$

Rough work / extra work

Current falling to 10 A

$I = 20 e^{-\lambda t}$, $\lambda = 2.303$. Find t when $I = 10$ A.

Solution: Current falling to 10 A

Step 1: 10 = 20 times e to the power (-λ t), so 1/2 equals that exponential

Step 2: Take natural log: ln(1/2) = -λ t

Step 3: t = ln 2 / λ. ln 2 = 2.303 log10(2)

Step 4: t = 2.303 log10(2) / 2.303 = log10(2)

t = 0.30 s

Loudness when intensity becomes I/4

L = 16 dB at intensity I. Intensity becomes I/4. Find the new loudness.

Solution: Loudness when intensity becomes I/4

Step 1: $L = 10 \log_{10}(I/I_0)$, $L' = 10 \log_{10}((I/4)/I_0)$

Step 2: $L - L' = 10 \log_{10}(4) = 10 \text{ times } 0.60 = 6$

Step 3: $L' = 16 - 6 = 10$

New loudness = 10 dB

Binomial when 1 is much larger than x

$$(1 + x)^n = 1 + n\,x \quad \text{if } |x| \ll 1$$

$$(1 - x)^n = 1 - n\,x$$

$$(1 + x)^{-n} = 1 - n\,x$$

Drop x^2 and higher powers only after the first term is 1 and |x| is much smaller than 1.

Rough work / extra work

$(1.007)^5$

Find $(1.007)^5$ using the small-x approximation.

Solution: $(1.007)^5$

Step 1: $1.007 = 1 + 0.007$, and $0.007 \ll 1$

Step 2: $(1 + 0.007)^5 = 1 + 5 \text{ times } 0.007 = 1 + 0.035$

1.035

Square root of 0.99

Find $\sqrt{(0.99)}$ using $(1 - x)^n$.

Solution: Square root of 0.99

Step 1: $0.99 = 1 - 0.01$, and the power is $1/2$

Step 2: $(1 - 0.01)^{1/2} = 1 - (1/2)(0.01) = 1 - 0.005$

0.995

Gravity at a small height

$g_h = GM/(R+h)^2$. If $h \ll R$, write a simple expression for g_h.

Solution: Gravity at a small height

Step 1: $R + h = R(1 + h/R)$, so $(R+h)^2 = R^2 (1 + h/R)^2$

Step 2: $g_h = (GM/R^2)$ times $(1 + h/R)^{-2}$

Step 3: $(1 + h/R)^{-2} = 1 - 2(h/R)$ because $h/R \ll 1$

$g_h = g (1 - 2h/R)$ where $g = GM/R^2$

Do not write the denominator as $(R + 2h)^2$. That is a different expansion.

Rough work / extra work

Trap: $(2.01)^3$ is not 2.03

Why is $(2.01)^3 = 2.03$ a wrong use of $(1+x)^n = 1 + n\,x$?

Solution: Trap: $(2.01)^3$ is not 2.03

Step 1: First write 2.01 = 2(1 + 0.005), not (1 + 0.01)

Step 2: The formula needs the leading 1. Here the leading number is 2.

$(2.01)^3 = 8\ (1.005)^3 = 8(1 + 0.015) = 8.12$

Physical quantity

A physical quantity can be measured. Love and hate cannot, so they are not physical quantities.

A measurable quantity is either a scalar or a vector.

Scalar: magnitude only. Vector: magnitude, direction, and triangle law.

Scalar and vector

	Scalar	Vector
Needs	magnitude only	magnitude and direction
Addition	ordinary + and -	triangle law
Sign	plus or minus is a value	plus or minus is a direction
Examples	mass, temperature, current, speed, energy	force, velocity, displacement, weight, electric field

Current has a conventional direction, but it does not follow the triangle law, so current is a scalar.

Rough work / extra work

Plus and minus on a vector

On a scalar, +10 and -10 differ in value. On a vector, the sign is the direction along a chosen line.

F1 = +20 N i-hat and F2 = -10 N i-hat

Same line, opposite senses. The unit vector i-hat carries the direction. The number is the magnitude with a sign.

How we write a vector

A arrow = A times A-hat

A is the magnitude (a positive number, with unit).

A-hat is the unit vector. It has magnitude 1 and no unit. It only points the direction.

Force is written F arrow, velocity v arrow, electric field E arrow. Mass, energy and speed stay without an arrow.

Arrow on paper

Tail to head. The head shows the direction. The length of the arrow shows the magnitude.

Two arrows of the same direction but different lengths are not equal.

Two arrows of the same length but different directions are not equal.

East is +i, west is -i, north is +j, south is -j, upward is +k.

A free vector may be slid parallel to itself. Magnitude and direction must stay the same. Turning it changes the vector.

Which one is a vector?

Which of the following is a vector? emf, surface tension, weight, focal length.

Solution: Which one is a vector?

Step 1: emf is a potential difference, a scalar

Step 2: surface tension = force per length, a scalar

Step 3: focal length is a length, a scalar

Step 4: weight = mg downward, so magnitude and direction

Weight is the vector.

Rough work / extra work

Which symbol is written correctly?

Work with an arrow, force without an arrow, or energy without an arrow?

Solution: Which symbol is written correctly?

Step 1: Work and energy are scalars, so no arrow

Step 2: Force is a vector, so it needs an arrow or bold type

Energy (E) is the correct writing among the three.

Direction does not automatically make a vector

A: no direction means scalar. B: a direction means vector. Which is correct?

Solution: Direction does not automatically make a vector

Step 1: A scalar really has no direction, so A is true

Step 2: Current has a direction but does not obey the triangle law

Step 3: So having a direction is not enough. B is false

A true, B false.

Angle between two vectors

Join tail with tail, or head with head. Take the smaller angle.

The bigger gap and the smaller gap add to 180 degrees only when the arrows are on crossing lines. The angle between the vectors is the smaller one, and it is never more than 180 degrees.

Do not measure the angle between a tail and a head when the arrows are placed carelessly. First bring tails together.

Rough work / extra work

Two arrows at 60 degrees, one reversed

Arrow A points right. Arrow B points down-right and the marked corner is 60 degrees. After reversing B onto the same tails, what is the angle between them?

Solution: Two arrows at 60 degrees, one reversed

Step 1: Place both tails at the same point

Step 2: The supplement of 60 degrees is 180 - 60 = 120

Angle between the vectors = 120 degrees.

Negative of a vector

-A has the same magnitude and the opposite direction.

If A = 10 i-hat, then -A = -10 i-hat.

On the paper, flip the head of the arrow. Do not change the length.

Unit vector of a 40 N force

A force of 40 N points north. What is the unit vector of that force?

Solution: Unit vector of a 40 N force

Step 1: F arrow = 40 N times (north unit vector)

Step 2: Unit vector = F arrow / magnitude

Step 3: The newton cancels. Direction does not carry a unit.

Unit vector = 1 north, not 1 N north.

Rough work / extra work

Types of vector — equality and lines

Equal: same magnitude, same direction, same kind of quantity.

Parallel: same direction. Magnitudes may differ. Every equal pair is parallel. A parallel pair need not be equal.

Negative: same magnitude, exactly opposite direction.

Anti-parallel: opposite direction, magnitudes may differ. Every negative pair is anti-parallel. An anti-parallel pair need not be a negative pair.

Types of vector — size and plane

Unit vector: magnitude 1, no unit, direction only.

Zero vector: magnitude 0. Example: velocity of a car at rest.

Collinear: all lie on the same straight line.

Coplanar: all lie in the same plane.

Orthogonal: angle between them is 90 degrees. Magnitudes may differ.

Concurrent: arrows act at the same point, even if magnitudes and directions differ.

Axial: direction is along the axis of rotation. Examples: torque, angular velocity, angular acceleration, angular momentum.

Vector 01 lock lines

log change of base is a ratio of logs in one common base

$(1+x)^n = 1 + n\,x$ only when the first term is 1 and $|x|$ is tiny

Vector = magnitude and direction, and it obeys the triangle law

A arrow = A times A-hat, and the angle is the smaller tail-to-tail angle

Practice

1. Redo the five log values without looking.

2. Redo both loudness questions and the decaying current.

3. Write five scalars and five vectors from memory.

4. Sketch equal, parallel, negative and orthogonal pairs.

Do not put an arrow on work or energy. Do not call current a vector.